

Cryptography

2 – Secret-key encryption: Block ciphers

Gabriel Chênevert

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JUNIA ISEN

Today

Malleability

Block ciphers

Construction

Modes of operation

Stream ciphers

Recall: A binary stream cipher is a symmetric cipher (E, D) with

$$E(k, x) = D(k, x) = x \oplus G(k)$$

where $G : \{0, 1\}^\ell \rightarrow \{0, 1\}^*$ is a CSPRNG.

Advantage: simplicity

Inconvenient: simplicity!

Example



Suppose Alice during a heated conversation sends to Bob:

$m = \text{no, I'm not angry!}$

encoded in ASCII then encrypted with Salsa20 using a 128-bit secret key.

Different attackers



Eve (a passive attacker): sees the ciphertext, learns nothing ✓



Oscar (an active attacker): is able to modify the ciphertext, may have a different goal !

Challenge



Find a way for Oscar (who doesn't know the secret key) to make it look like Alice is shouting by making sure Bob receives the message

$$m' = \text{NO, I'M NOT ANGRY!}$$

Hints

- the ASCII codes for upper- and lower-case letters differ by exactly 1 bit

ex.: $A = 41 = 01000001$, $a = 61 = 01100001$

NB: $\text{Space} = 20 = 00100000$ thus $A = a \oplus \text{Space} !$

- Malleability:**

$$E(k, m \oplus y) = E(k, m) \oplus y$$

$$D(k, c \oplus y) = D(k, c) \oplus y$$

Can be seen as a weakness – or a feature! " $\Delta c = \Delta m$ "

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New point of view

Consider (E, D) a symmetric cipher with $\mathcal{M} = \mathcal{C}$.

For given $k \in \mathcal{K}$,

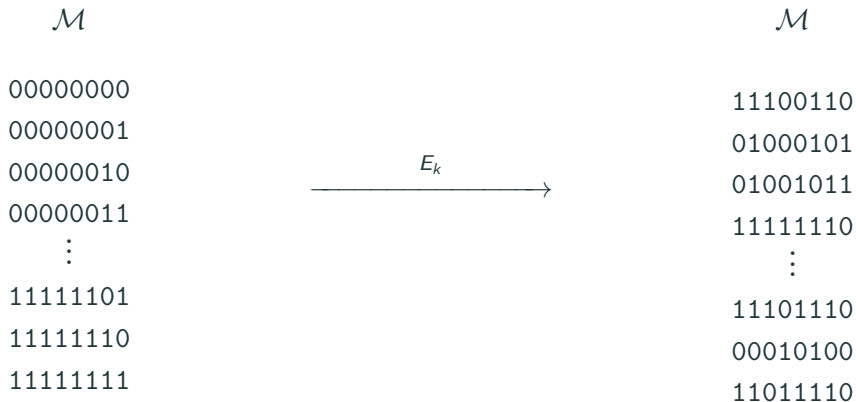
$$E_k := E(k, \cdot) : \mathcal{M} \longrightarrow \mathcal{M}$$

admits $D_k := D(k, \cdot)$ as inverse

hence E_k is a **permutation** of $\mathcal{M}$ (bijection from $\mathcal{M}$ to $\mathcal{M}$)

E_k as a permutation

e.g. with $|\mathcal{M}| = 2^8$:



Idea

E_k should be thought of as a *pseudo-random permutation* of $\mathcal{M}$.

In practice: undistinguishable from a random *function* $\mathcal{M} \rightarrow \mathcal{M}$.

Allows one to:

- reuse keys (with some care!)
- work with small messages (**blocks**)

Note: typically $|\mathcal{K}| \ll |\text{permutations of } \mathcal{M}| = |\mathcal{M}|! \approx |\mathcal{M}|^{|\mathcal{M}|}$

ex.: $|\mathcal{K}| = |\mathcal{M}| = 2^{128}$, $|\mathcal{S}_{\mathcal{M}}| \approx |\mathcal{M}|^{|\mathcal{M}|} \approx 2^{43556142965880123323311949751266331066368}$ (!)

But I liked stream ciphers!

Block ciphers are typically slower, but

- can avoid linearity (malleability) problems;
- are much more versatile!

For example: you can get a *parallelizable* CSPRNG for free from a block cipher E

$$G(k) := E(k, 0) \parallel E(k, 1) \parallel E(k, 2) \parallel \dots$$

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Do these things actually exist?

Shannon's paradigm: *confusion* and *diffusion* (stream ciphers miss the diffusion part)

Essentially all modern examples use an iterative design where the plaintext is encrypted a certain number of times by a **round function** performing (a small amount of) confusion and diffusion

$$\begin{cases} x_0 = m, \\ x_{i+1} = R(k_{i+1}, x_i), & 0 \leq i < r \\ E(k, m) = x_r \end{cases}$$

preceded by a **key scheduling** process $k \mapsto (k_1, \dots, k_r)$.

Famous examples

n -bit block, ℓ -bit key, r rounds

- **Lucifer** (IBM, 1971) $n = \ell = 128$, $r = 16$
- **Data Encryption Standard** (NIST, 1977) $n = 64$, $\ell = 56$, $r = 16$

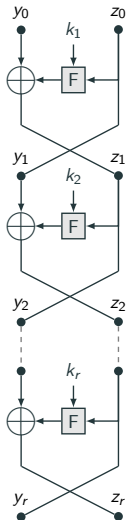
Successful brute force attack in 1997!

Survived for a while in the form of Triple DES in legacy hardware/software

- **Rijndael** (KU Leuven, 1998) *aka* **Advanced Encryption Standard** (NIST, 2001)
 $n = 128$, $\ell \in \{128, 192, 256\}$, $r \in \{10, 12, 14\}$.

But also: RC5/RC6, IDEA, Serpent, Blowfish/Twofish, ...

Design of DES



16-round **Feistel network** :

Write each $x_i = y_i \parallel z_i$ left and right parts

Round function:

$$\begin{cases} y_{i+1} = z_i \\ z_{i+1} = y_i \oplus F(k_{i+1}, z_i) \end{cases}$$

Easy to implement in hardware
(and invert – exercise!)

Security proof

Theorem (Luby-Rackhoff, 1988)

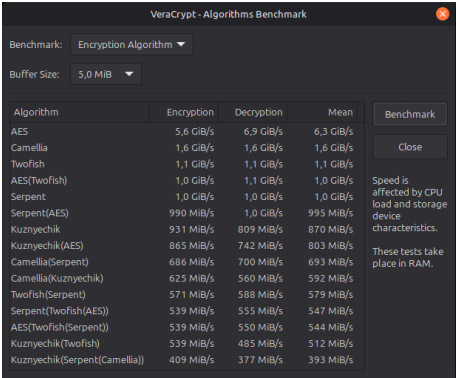
Three rounds of a Feistel network with inner function F a CSPRNG using k as a seed is computationally undistinguishable from a random permutation.

In practice: increase the number of rounds to take into account the fact that F might not be a provably good CSPRNG.

Still: the original DES can now be broken by exhaustive key search in a couple of hours (minutes?) with dedicated hardware (e.g. **COPACOBANA**)

Today

In practice: **use AES** or some other NIST finalist / international alternative



Algorithm	Encryption	Decryption	Mean
AES	5,6 GiB/s	6,9 GiB/s	6,3 GiB/s
Camellia	1,6 GiB/s	1,6 GiB/s	1,6 GiB/s
Twofish	1,1 GiB/s	1,1 GiB/s	1,1 GiB/s
AES(Twofish)	1,0 GiB/s	1,1 GiB/s	1,0 GiB/s
Serpent	1,0 GiB/s	1,0 GiB/s	1,0 GiB/s
Serpent(AES)	990 MiB/s	1,0 GiB/s	995 MiB/s
Kuznyechik	931 MiB/s	809 MiB/s	870 MiB/s
Kuznyechik(AES)	865 MiB/s	742 MiB/s	803 MiB/s
Camellia(Serpent)	686 MiB/s	700 MiB/s	693 MiB/s
Camellia(Kuznyechik)	625 MiB/s	560 MiB/s	592 MiB/s
Twofish(Serpent)	571 MiB/s	588 MiB/s	579 MiB/s
Serpent(Twofish(AES))	539 MiB/s	555 MiB/s	547 MiB/s
AES(Twofish(Serpent))	539 MiB/s	550 MiB/s	544 MiB/s
Kuznyechik(Twofish)	539 MiB/s	485 MiB/s	512 MiB/s
Kuznyechik(Serpent(Camellia))	409 MiB/s	377 MiB/s	393 MiB/s

Speed is affected by CPU load and storage device characteristics.
These tests take place in RAM.

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Now suppose the message to be encrypted is longer than a single block:

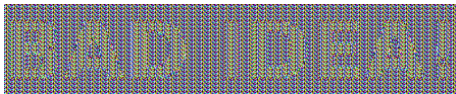
$$m = m_1 \parallel m_2 \parallel m_3 \parallel \dots$$

How to use a block cipher (e.g. AES) to encrypt m ?

- **Electronic Code Book (ECB) mode:**

$$\begin{cases} c_i = E(k, m_i) \\ m_i = D(k, c_i) \end{cases}$$

ECB mode?



Problem: equal blocks yield equal ciphertexts

Should use (pseudo-) **probabilistic encryption**:

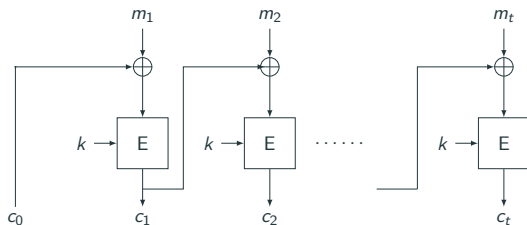
a given block shouldn't always have the same encryption

~> use of either *random value* or *nonce* (counter)

Let's look at two of the simplest ways to do it:

Cipher Block Chaining (CBC) mode

$$\begin{cases} c_0 = \text{random Initial Value} \\ c_i = E(k, m_i \oplus c_{i-1}) \end{cases}$$



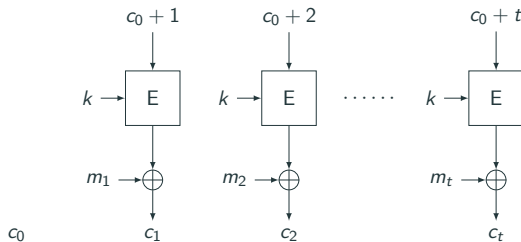
- Encryption is sequential (but decryption can be parallelized)

$$m_i = D(k, c_i) \oplus c_{i-1}$$

- Message has to be padded to a multiple of the block length
- Crucial that random IV is non-predictable (chosen plaintext attack)

Randomized counter (CTR) mode

$$\begin{cases} c_0 = \text{random IV} \\ c_i = m_i \oplus E(k, c_0 + i) \end{cases}$$



- Block cipher is effectively turned into a stream cipher
- No padding problem
- Highly parallelizable
- Random IV prevents reuse of key stream

Other modes

Many other modes that achieve specific goals exist.

- feedback modes: CFB, OFB, ...
- device encryption: LRW, XEX, XTS, ...
- authenticated encryption: OCB, EAX, GCM, ...

(more on that next time)